

DeepSeek-VL-1.3B (Featherweight)

This model is the "**Speed King**" of the list. With only 1.3 billion parameters, it is incredibly small and fast, designed to work with minimal resources.

- **Why it Matters:** Usually, models of this tiny size can identify objects but cannot understand context. DeepSeek-VL is unique because it can perform **basic logic**, unlike competitors such as Moondream.
- **Role in Project:** It is perfect for tasks where speed is critical. Use it for quick filtering or simple checks (e.g., "Is there a car in this photo?") before passing the job to a larger model.

Qwen2-VL-2B (Featherweight)

Even though it is small, this model serves as the "**Main Pilot**." It offers the best balance between speed and intelligence for general daily tasks.

- **Why it Matters:** The Qwen2 architecture is very modern. It handles video analysis and supports multiple languages better than older models. It is light enough to run on mobile devices but smart enough to understand user questions.
 - **Role in Project:** This is the default engine. It handles standard user queries efficiently, saving the heavy lifting for the larger models only when necessary.
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LLaVA-Phi-3-Mini (Lightweight)

This model represents the "**Legacy Test**." It combines the classic, industry-standard LLaVA architecture with Microsoft's highly efficient Phi-3 "brain."

- **Why it Matters:** It proves that you don't need a massive model to get good results. By using Microsoft's specialized training, this model punches above its weight class, especially when reading text inside images (OCR).
- **Role in Project:** It is used to test how well older visual methods work with modern language processors. It is a reliable option for reading documents or signs.

Phi-3.5 Vision (Lightweight)

This is the updated version of the previous model and serves as the "**Instruction Expert**." It is specifically tuned to follow orders precisely.

- **Why it Matters:** While some AI models "hallucinate" (make things up), Phi-3.5 is excellent at sticking to strict rules. It excels at understanding "If-Then" logic (e.g., "If the screen is red, tell me to stop").
 - **Role in Project:** Ideal for workflows that require strict adherence to guidelines, such as filling out forms, safety checks, or following multi-step commands without error.
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DeepSeek-Janus-7B (Middleweight)

This model is the "**Innovator.**" DeepSeek uses a unique internal structure that separates visual understanding from image generation, offering a different perspective than the Qwen series.

- **Why it Matters:** Most models follow a standard path. Janus tries a new architecture that allows for more flexible visual processing. It is a strong competitor that provides alternative answers when standard models struggle.
- **Role in Project:** Use this when you need a "second opinion." If the other models are confused by an abstract or unusual image, Janus might solve it.

Qwen2-VL-7B (Middleweight)

This model is the "**Vision Specialist.**" Its superpower is a technology called **NaViT** (Native Resolution Vision Transformers).

- **Why it Matters:** Most AI models shrink or crop an image into a square before looking at it, which ruins small details. Qwen2-VL sees the image in its original shape and resolution. It does not miss the small stuff.
 - **Role in Project:** The go-to model for details. Use it for analyzing complex diagrams, reading tiny text in the background, or spotting small defects that other models might miss.
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LLaVA-Mistral (Heavyweight)

This is "**The Standard.**" It combines the reliable LLaVA visual system with Mistral 7B, which is one of the most respected open-source language models.

- **Why it Matters:** Mistral is known for speaking very naturally and fluently. It provides stable, consistent, and high-quality descriptions. It is the benchmark against which other models are measured.
- **Role in Project:** It provides the most reliable general-purpose descriptions. When you need a solid, human-like explanation of an image without taking risks, this is the choice.

LLaVA-Llama-3 (Heavyweight)

This model is "**The Brain.**" Powered by Meta's Llama-3, it offers the highest level of linguistic reasoning and safety awareness in your list.

- **Why it Matters:** It doesn't just "see" pixels; it understands the deeper meaning. It has superior logic capabilities and has been heavily trained on safety, making it the best at detecting harmful or sensitive content.
- **Role in Project:** Reserved for the hardest tasks. Use it for deep reasoning, analyzing the intent behind an image, or creating safe, detailed reports for sensitive topics.